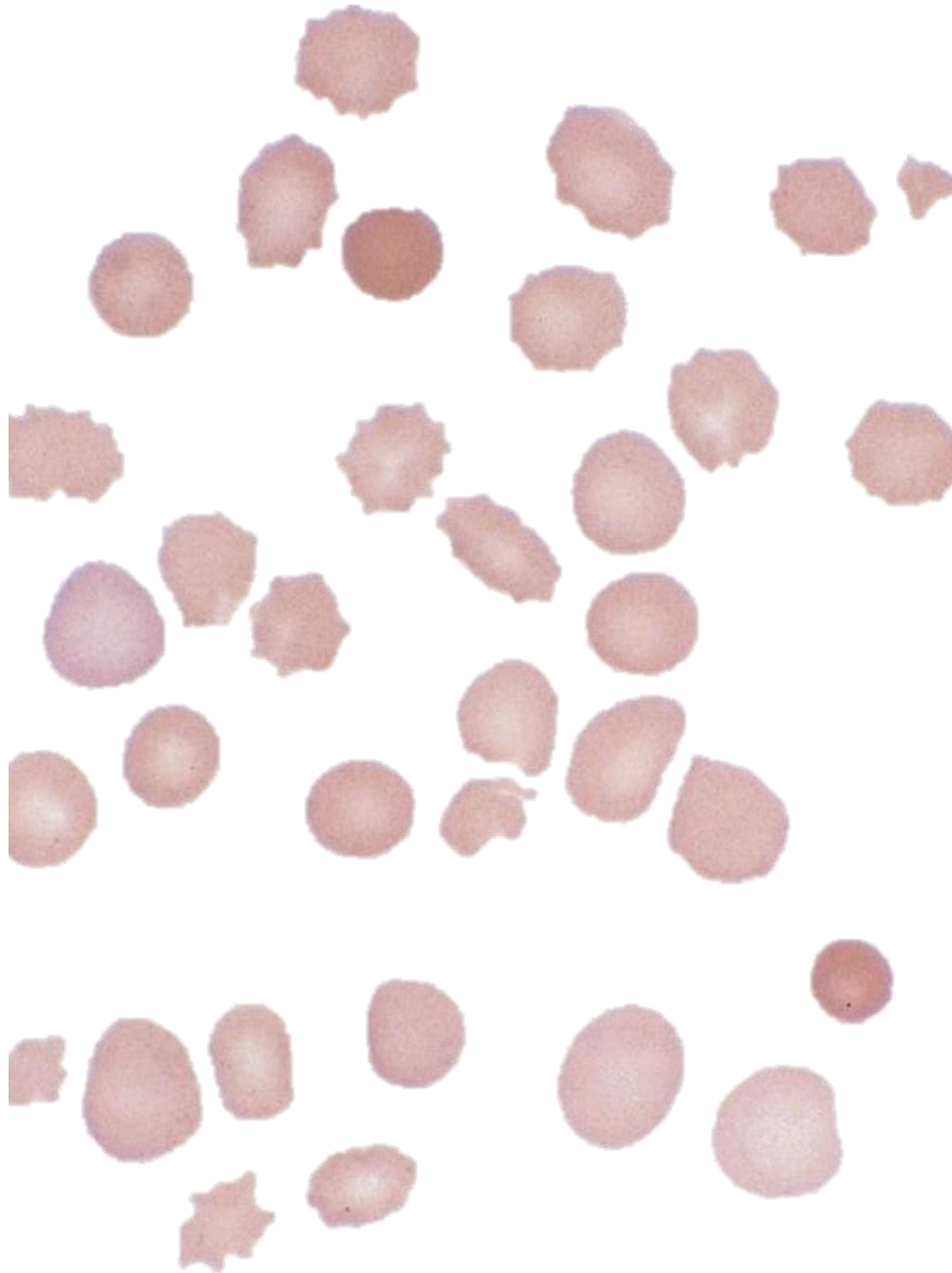
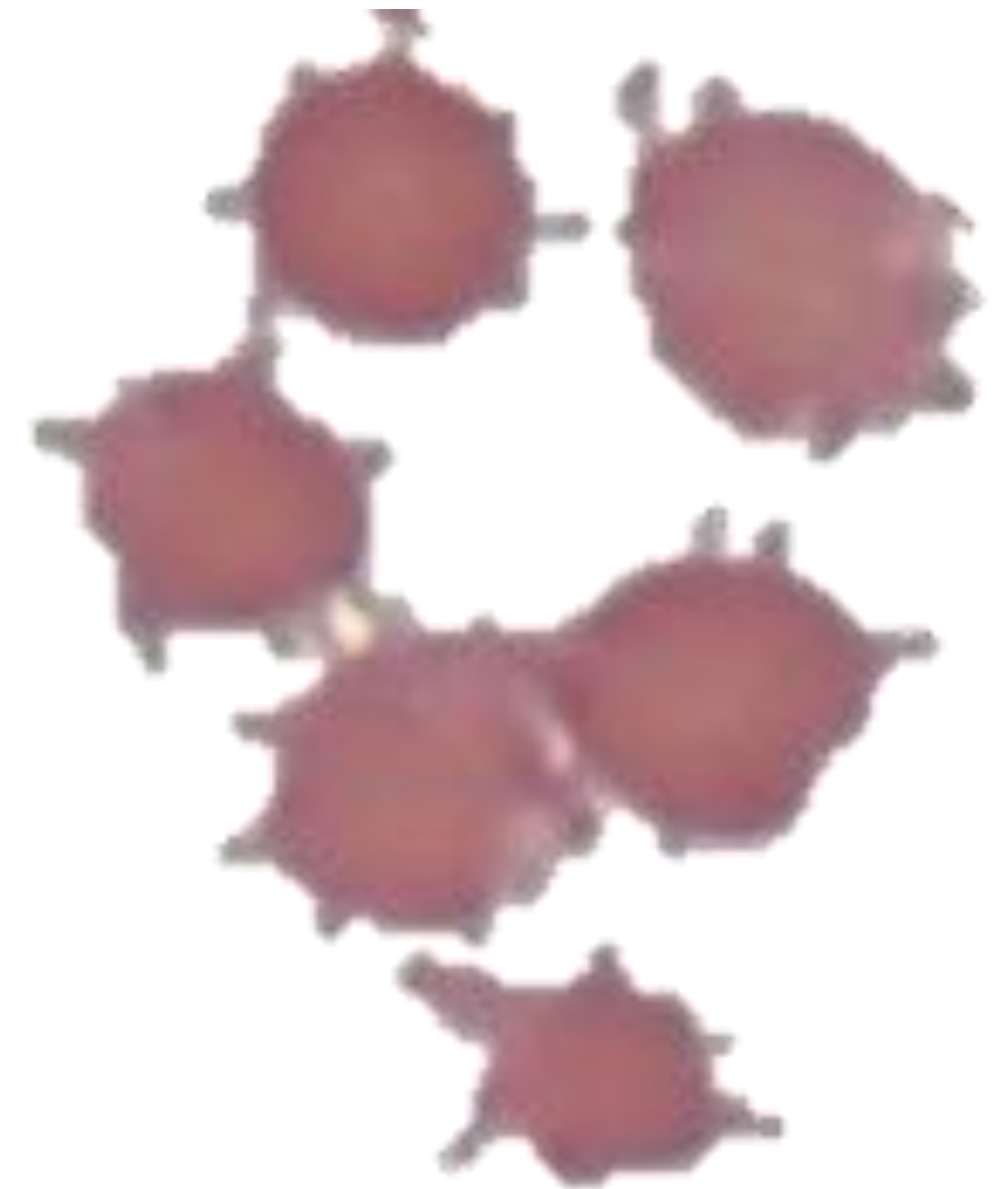




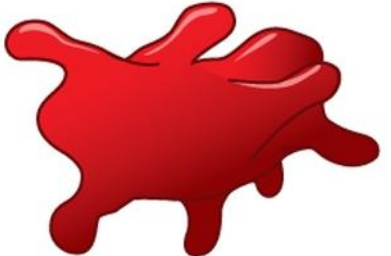
Poikilocytosis

- Term used to describe nonspecific variation in RBC shape
- Some shapes and sizes
 - Characteristic of specific underlying hematological disorder or malignancies
 - eg schistocytes, echinocytes, spherocytes
- Some morphology may be artifactual
 - eg ovalocytes, “target” (plt on top of RBC)
 - Poorly made or improperly stained smears

Acanthocyte (Spur cell)

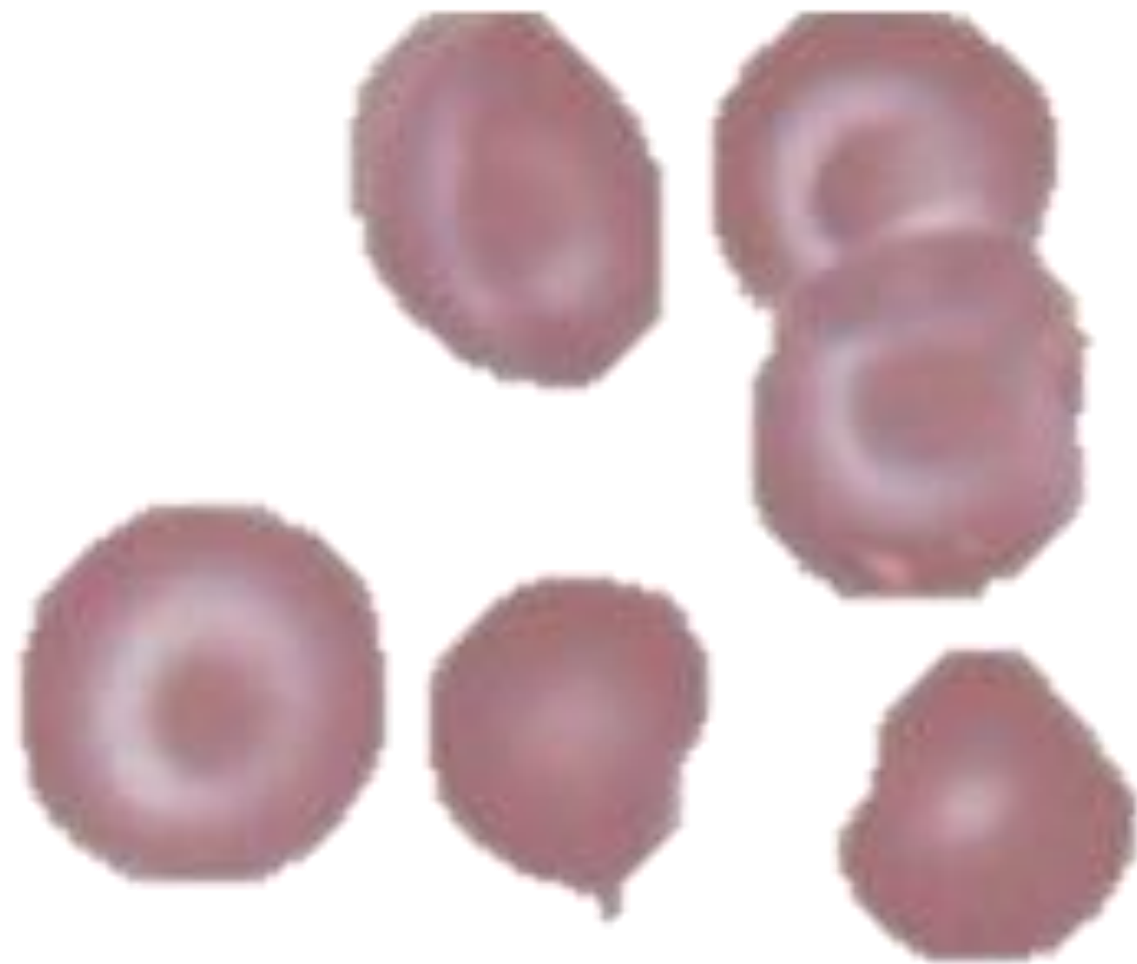


- Small spherical cells (do not have central pallor) with irregular thornlike projections
 - Projections may have small bulblike tips
- RBC membranes have altered lipid content
 - ↑ free cholesterol accumulation leading to ↑ OM fluidity

Terminology	Synonyms	Description	Associated Disease States	Drawing of Morphology
Acanthocyte (spike)	Spur cell	Red cells with spicules of varying length irregularly distributed over the surface; no area of pallor	Abetalipoproteinemia; alcoholic liver disease; disorders of lipid metabolism; post splenectomy; fat malabsorption; retinitis pigmentosa	

Target Cells (Codocytes)

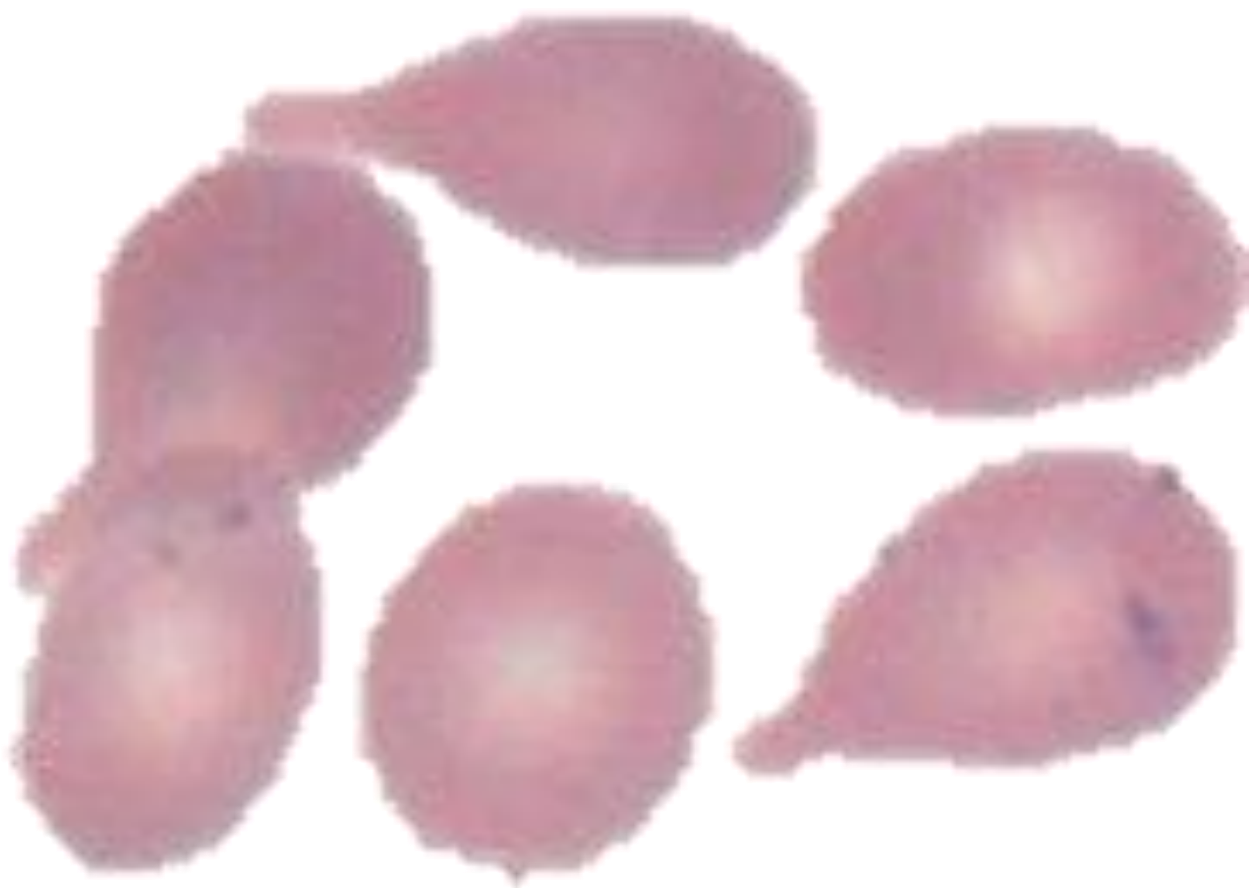
- Thin, bell-shaped cells
 - ↑ surface-to-volume ratio
 - On PBS look like a target cell with a bull's eye in the center
 - Bull's eye is surrounded by an achromic zone and a thin outer ring of pink-staining Hgb



Terminology	Synonyms	Description	Associated Disease States	Drawing of Morphology
Codocyte (bell)	Target cell	Thin, bell-shaped, with increased surface-to-volume ratio; on stained blood smears, appears as a target with a central bull's-eye, surrounded by achromic zone and outer ring of hemoglobin	Hemoglobinopathies; thalassemias; obstructive liver disease; iron deficiency anemia; splenectomy; renal disease; LCAT deficiency	

Teardrop Cells (Dacryocytes)

- RBCs that are elongated on one end to form a teardrop
 - Formed when RBCs containing inclusions traverse the spleen
- The area with the inclusion has difficulty passing through the splenic filter
- Cell is stretched, cannot return to its normal shape



Terminology	Synonyms	Description	Associated Disease States	Drawing of Morphology
Dacryocyte (tear)	Teardrop	Round cell with a single elongated or pointed extremity; may be microcytic and/or hypochromic	Myelophthisic anemias; primary myelofibrosis (PMF); thalassemias	

Sickle Cells (Drepanocytes)

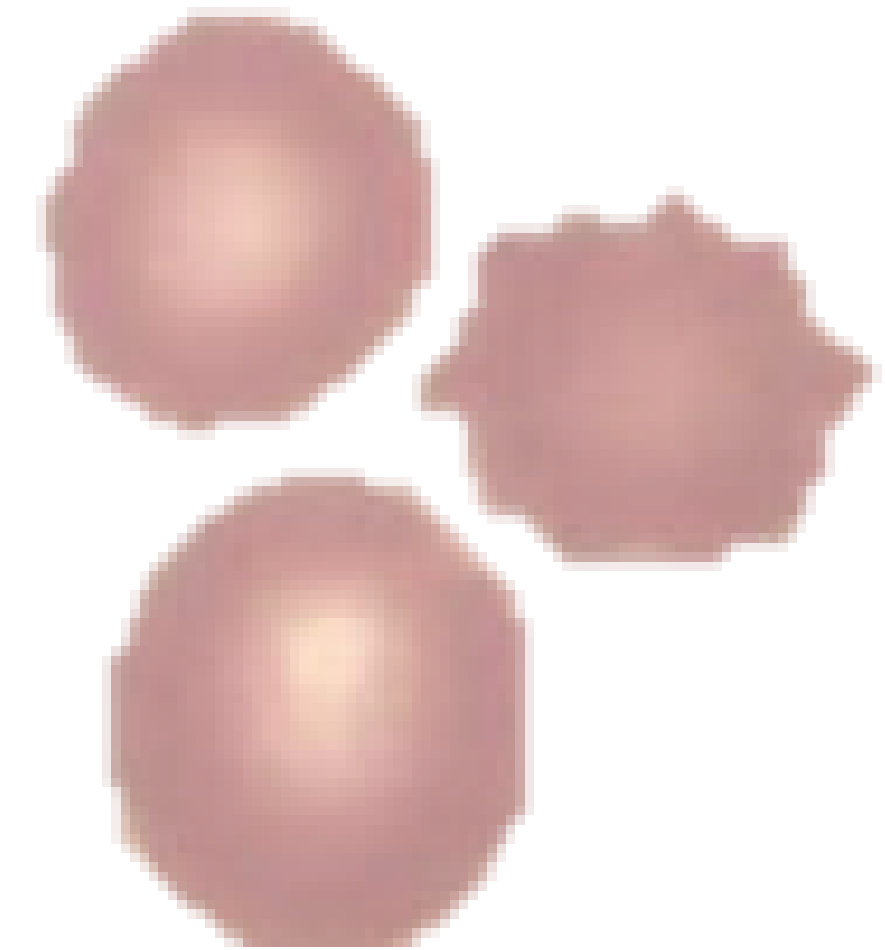


- Elongated, crescent-shaped RBC with pointed ends
 - Reversible forms have round ends with a flat side
- Abnormal HGB SS polymerizes into rods in ↓ O₂ tension or ↓ pH conditions
 - 1st, transforms into a holly leaf shape



Terminology	Synonyms	Description	Associated Disease States	Drawing of Morphology
Drepanocyte (sickle)	Sickle cell	Contain polymerized hemoglobin showing various shapes: sickle, crescent, or boat shaped	Sickle cell disorders	

Echinocytes (Burr cells/Crenated)

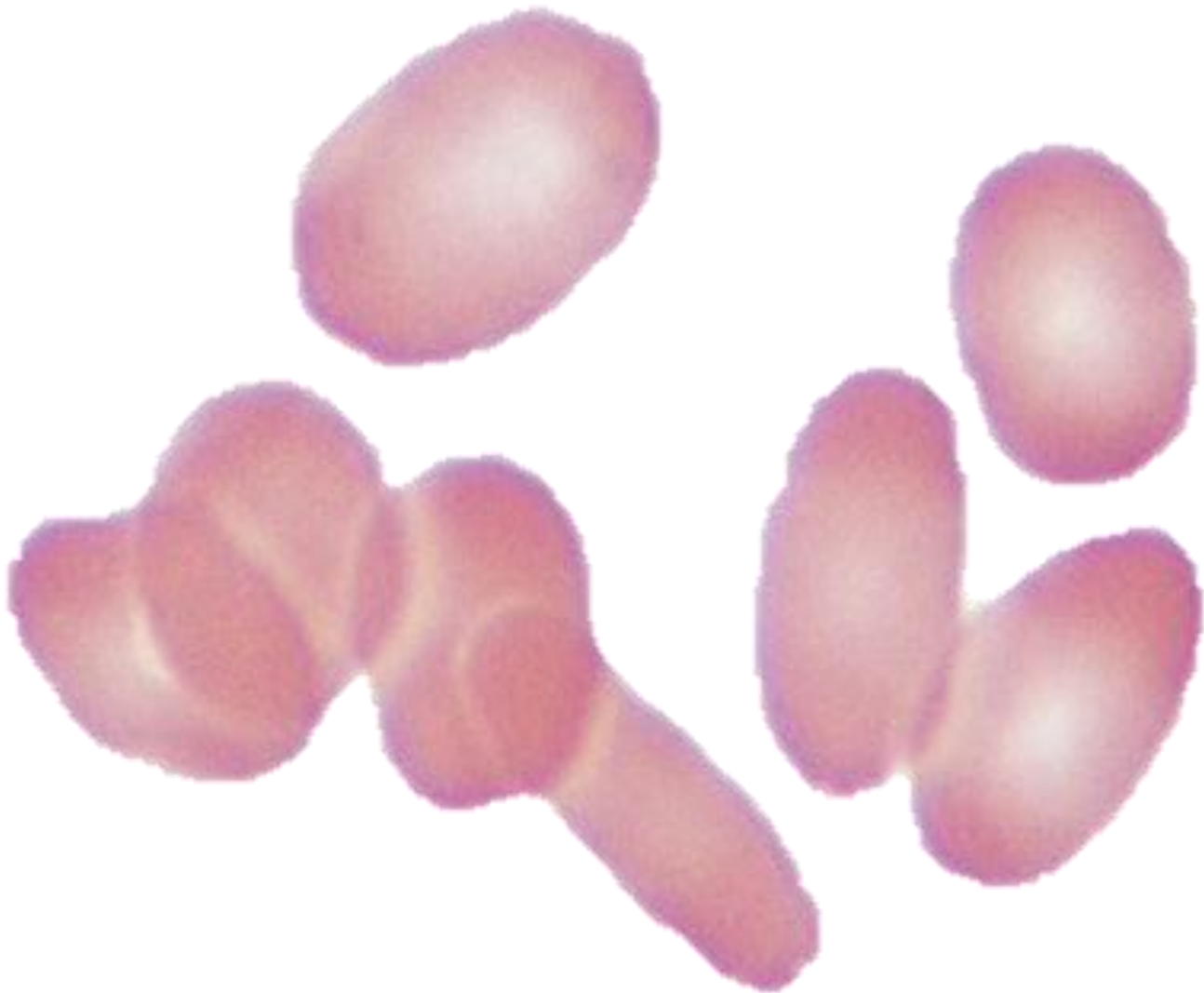


- Smaller than normal RBCs with regular, spinelike projections
 - Is reversible (can revert to a discocyte)
 - Can become a spherocyte
 - Due to removal of the membrane spines in spleen
 - Common artifact in stained blood smears
 - True shape = $\uparrow$ in the area of the outer membrane leaflet

Terminology	Synonyms	Description	Associated Disease States	Drawing of Morphology
Echinocyte (sea urchin)	Burr cell; crenated cell	Spiculated red cells with short equally spaced projections over the entire surface	Liver disease; uremia; pyruvate kinase deficiency; peptic ulcers; cancer of stomach; heparin therapy	

Ovalocytes/Elliptocytes

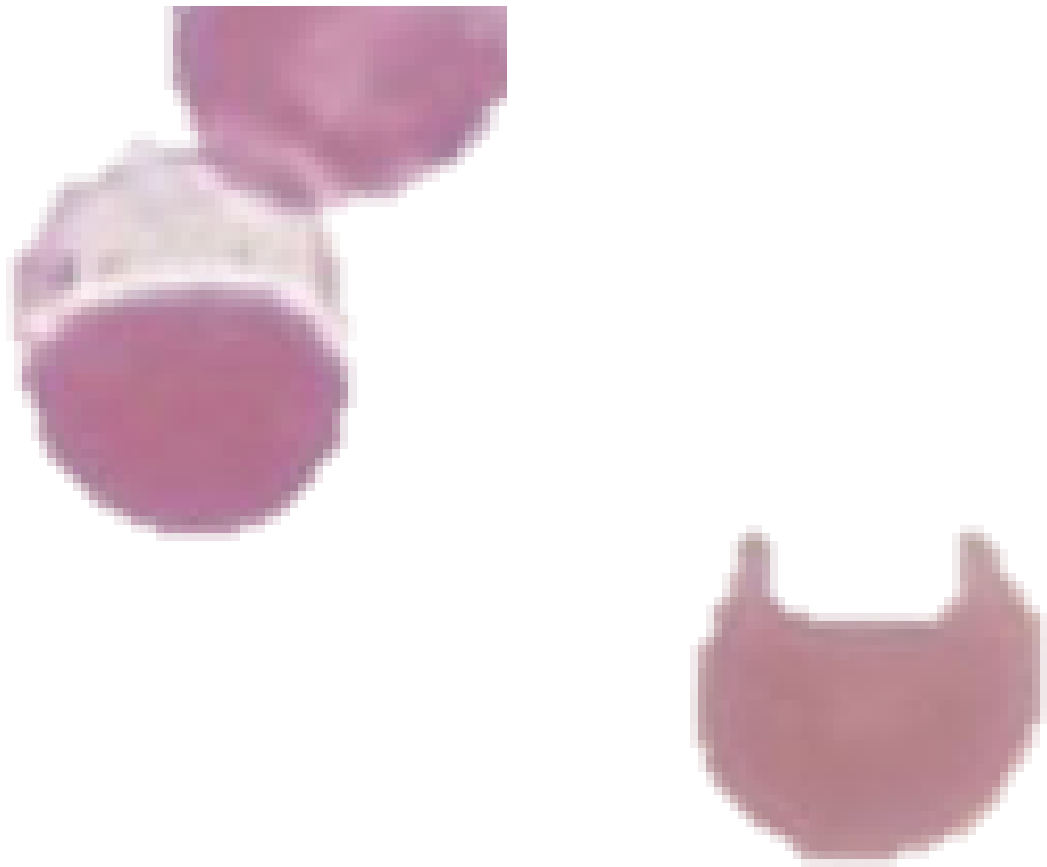
- Also called *pencil* or *cigar* cells
 - Vary from elongated ovals to elongated rodlike cells
 - Has central pallor
 - Formed from alterations in RBC membrane skeleton and flow through vessels



Terminology	Synonyms	Description	Associated Disease States	Drawing of Morphology
Elliptocyte (oval)	Ovalocyte; pencil cell; cigar cell	Oval to elongated ellipsoid cell with central area of pallor and hemoglobin at both ends	Hereditary elliptocytosis; iron deficiency anemia; thalassemia; anemia associated with leukemia	

Keratocytes or Helmet Cells

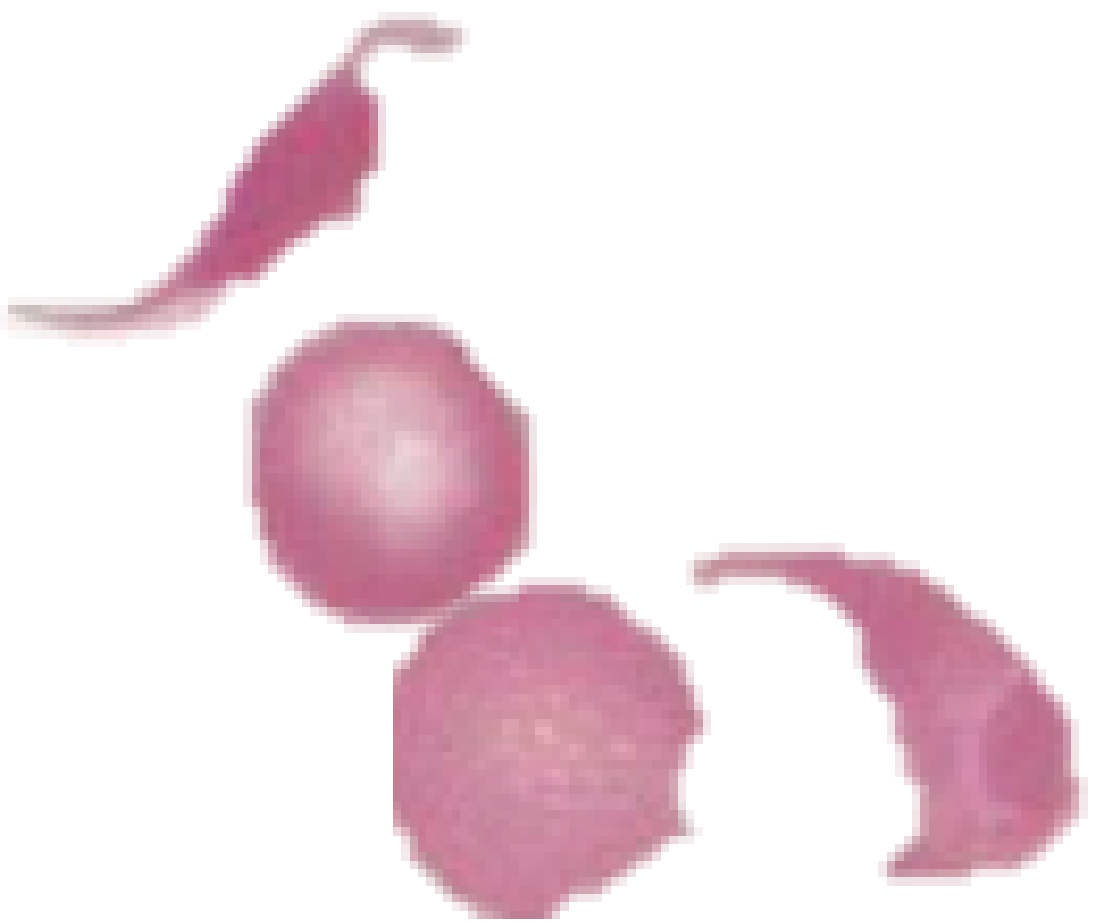
- One side concavity and two hornlike protrusions
- Produced by impalement on a fibrin strand
 - The two halves of the RBC hang over the strand
 - The membranes touch and fuse
 - Produces a vacuole-like inclusion on one side called blister cell
 - The vacuole bursts, leaving a notch with two spicules

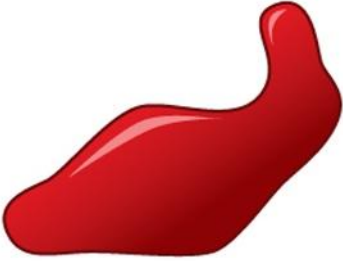


Terminology	Synonyms	Description	Associated Disease States	Drawing of Morphology
Keratocyte (horn)	Helmet cell; horn-shaped cell	Red cells with one or several notches with projections that look like horns on either end	Microangiopathic hemolytic anemias; heart-valve hemolysis; Heinz-body hemolytic anemia; glomerulonephritis; cavernous hemangiomas	

Schistocytes (RBC Fragments)

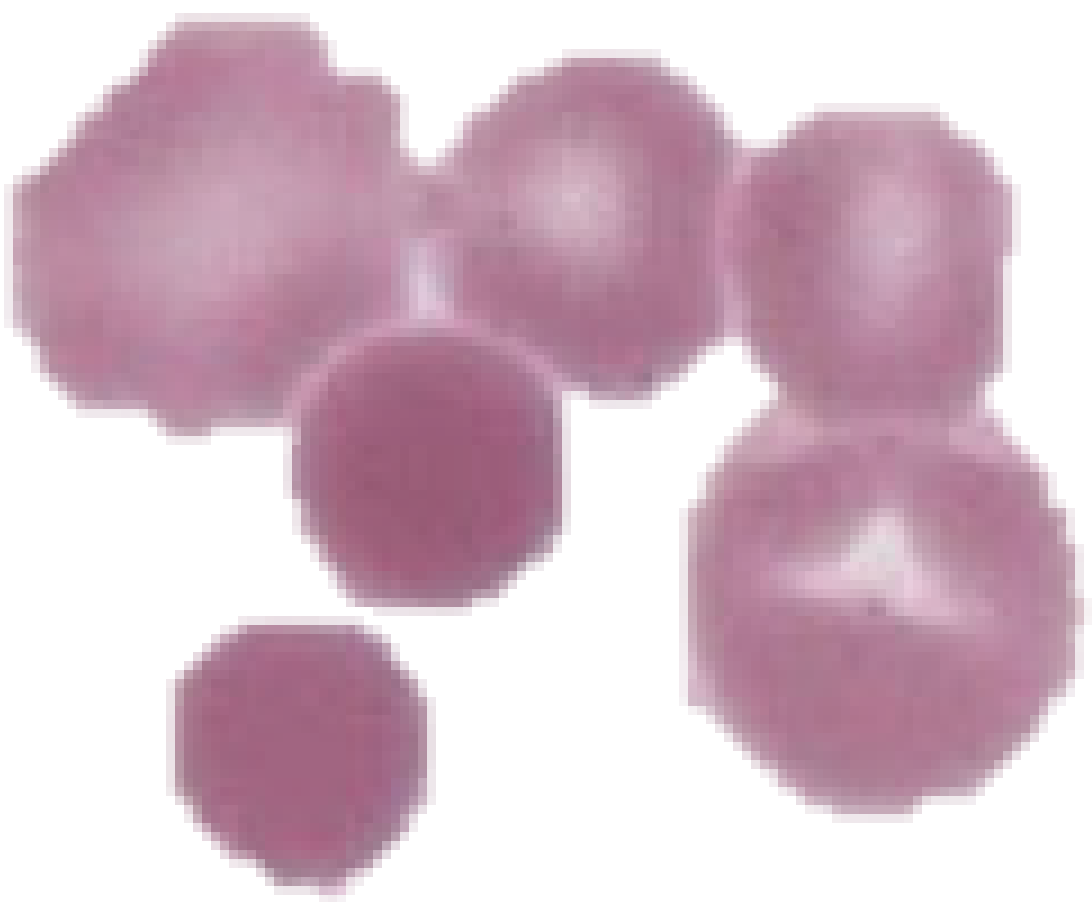
- Caused by mechanical damage to cells
 - Variety of shapes
 - Triangular, comma shaped, helmet
 - Usually microcytic
 - Normal deformability but reduced survival
 - Fragments may hemolyze or be culled

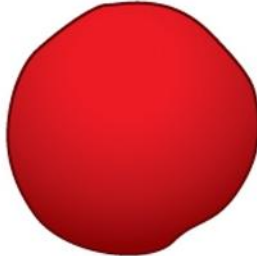


Terminology	Synonyms	Description	Associated Disease States	Drawing of Morphology
Schistocyte (cut)	Schizocyte; fragmented cell	Fragments of red cells; variety of shapes including triangles, commas; microcytic	Microangiopathic hemolytic anemias; heart-valve hemolysis; disseminated intravascular coagulation; severe burns; uremia	

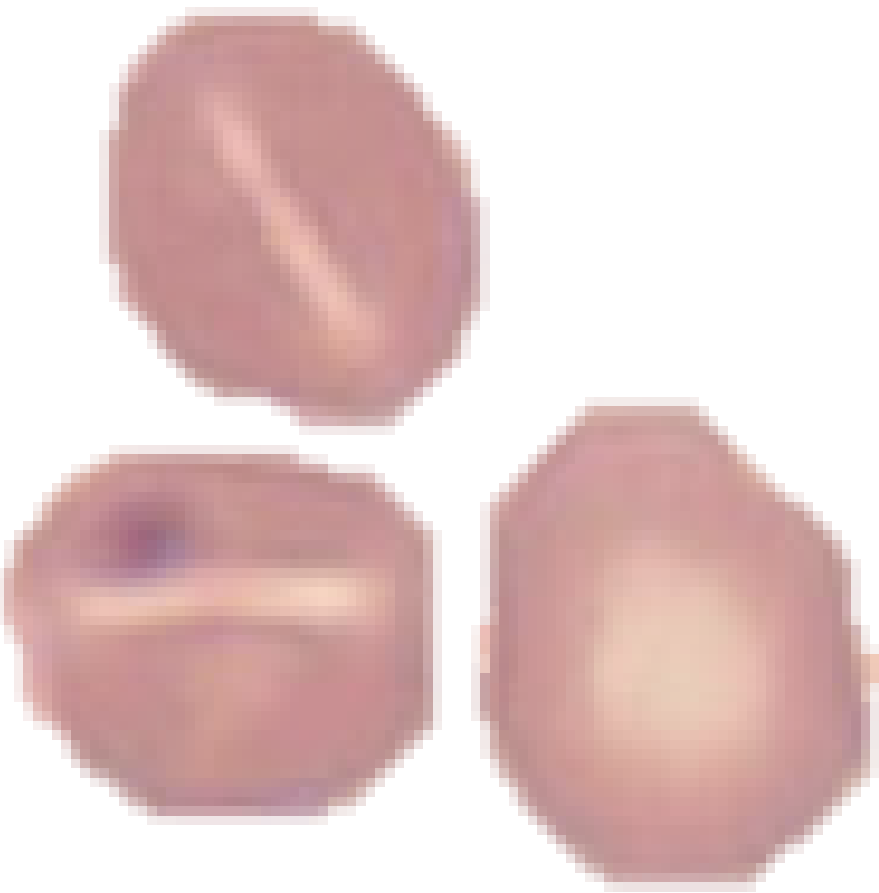
Spherocytes

- Biconcavity lost due to ↓ surface to volume ratio
 - Densely stained sphere lacking central pallor
 - Appear microcytic but volume (MCV) is usually normal
 - Only poikilocyte classified as “hyperchromic”



Terminology	Synonyms	Description	Associated Disease States	Drawing of Morphology
Spherocyte	—	Spherocytic red cells with dense hemoglobin content (hyperchromatic); lack an area of central pallor	Hereditary spherocytosis; immune hemolytic anemias; severe burns; transfusion with ABO incompatibility; Heinz-body hemolytic anemias	

Stomatocytes (Mouth/Clam Cells)



- Slit like area of pallor
 - Shape is reversible
 - True shape = $\uparrow$ in lipid content or area of inner leaflet
- Can appear as artifact on smear
 - Care should be taken when calling stomatocytes

Terminology	Synonyms	Description	Associated Disease States	Drawing of Morphology
Stomatocyte (mouth)	Mouth cell; cup form; mushroom cap	Uniconcave red cells with the shape of a very thick cup; on stained blood smears cells have an oval or slitlike area of central pallor	Hereditary stomatocytosis; spherocytosis; alcoholic cirrhosis; anemia associated with Rh null disease; lead intoxication; neoplasms	